

Short-term earthquake precursor pattern detection in the Eastern Indian Plate margin: a leakage-controlled machine-learning framework benchmarked against clustering persistence.

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Highlights

- Embargoed time-series CV closes window-overlap and label look-ahead leakage.
- Post-1995 random forest attains ROC-AUC 0.89 and recall 0.80 at 11% false alarms.
- A parameter-free Omori kernel outperforms every learned model (ROC-AUC 0.933).
- Event-level audit: 78% of $M_w \geq 4.5$ mainshocks had no scoreable local sequence.
- Hard physical gating fails through conjunction with an uninformative feature.

Abstract

Machine-learning studies of short-term earthquake forecasting are routinely weakened by validation leakage, catalog non-stationarity, arbitrary decision thresholds, and the absence of clustering benchmarks. We address all four in a single reproducible framework applied to the Eastern Indian Plate margin (10–30°N, 85–98°E). A 55-year reviewed USGS catalog (7,188 events) is converted into 7,168 overlapping 20-event sequences described by thirteen physics-informed features and labelled positive when a Moment Magnitude scale (M_w) ≥ 4.5 event follows within 15 days and 100 km. A three-seed random forest is evaluated under ten-fold forward-chaining cross-validation with a 20-sequence embargo that purges both window-overlap and label look-ahead leakage, restricted primarily to the post-1995 completeness era. The forest reaches ROC-AUC 0.892 and PR-AUC 0.836 at 24.8% prevalence; Pareto-knee thresholding gives recall 0.796 (95% CI 0.773–0.817) at a false alarm rate of 0.098. Yet a parameter-free Omori-kernel persistence baseline, evaluated on the identical folds, attains ROC-AUC 0.933 and recall 0.866 at FAR 0.083, and no learned model or ensemble improves on it. Stratified analysis shows the forest carries essentially no skill (ROC-AUC 0.49) outside clustered periods. The two test windows that defeat the forest are quiescent intervals dominated by deep, spatially isolated Indo-Burman intraslab events, and the same windows defeat the Omori kernel to an equal degree. An event-level audit of all 2,363 $M_w \geq 4.5$ events in the evaluation period makes the ceiling concrete: alerts precede 84.3% of events for which any local sequence existed, but 78% of events, including every event deeper than 70 km, presented no scoreable local sequence, so unconditional per-event detection is 18.6% with a median lead of 1.5 days. We conclude that, as commonly labelled, this detection task measures clustering persistence rather than precursory anomaly, quantify exactly where the unsolved problem lives, and release the complete seeded pipeline so that both claims can be verified line by line.

Keywords: earthquake precursor; short-term forecasting; random forest; Omori clustering; multi-objective threshold selection; catalog completeness; time-series cross-validation; Eastern Indian Plate

1. Introduction

The Eastern Indian Plate margin, encompassing the eastern Himalayan syntaxis, the Shillong Plateau, the Bengal Basin, and the Indo-Burman subduction wedge, accommodates oblique India–Eurasia convergence through a network of thrust and strike-slip systems that have produced great earthquakes including the 1897 Shillong ($M_w \approx 8.1$) and 1950 Assam ($M_w \approx 8.6$) events [1–3]. Geodetic evidence indicates that a locked megathrust beneath densely populated Bangladesh is accumulating strain sufficient for future great earthquakes [4]. With more than 200 million people living on deep sediments that amplify ground motion, even moderate $M_w \geq 4.5$ events carry engineering significance for this region, and any statistically defensible short-term situational-awareness capability has direct value for hazard operations.

Machine learning has re-energised earthquake forecasting research, from aftershock-location modelling with deep networks [5] through meta-analyses of neural-network prediction studies [6] to community roadmaps for ML-based forecasting [7,8]. Regional pattern-recognition studies have reported encouraging skill using seismicity-derived features [9–11]. Yet several methodological weaknesses recur. Random or shuffled cross-validation lets information travel backwards in time, and the purged, embargoed validation designs developed for overlapping financial series [12] are almost never applied even though overlapping sequence windows create exactly the same leakage structure. Multi-decadal catalogs are treated as homogeneous despite order-of-magnitude changes in detection capability that distort every rate-based feature [13,14]. Classifier outputs are cut at an arbitrary 0.5 despite grossly asymmetric error costs. Pooled metrics conceal temporal instability. Most seriously, reported skill is rarely referenced to what trivial earthquake clustering already explains, a criticism raised forcefully by the meta-analytic literature [6] and central to modern forecast evaluation practice [15].

This paper addresses these weaknesses jointly and reports what happens when they are all closed at once. The contributions are: (i) an event-driven micro-sequence representation with thirteen physics-informed features, including two bounded convexity measures of Benioff-strain and event-arrival acceleration that avoid the fitting instabilities of classical accelerating-moment-release analysis [16–18]; (ii) an embargoed time-series cross-validation protocol whose 20-sequence gap purges window-overlap and label look-ahead leakage simultaneously, with the inflation from unembargoed validation quantified; (iii) a dual completeness protocol that treats catalog-era heterogeneity as an explicit experimental variable; (iv) operational alert construction by fuzzy Pareto-knee thresholding, extended by a physical plausibility gate family whose failure we dissect quantitatively; and (v) a parameter-free Omori persistence benchmark evaluated on identical folds, which reframes the headline result: the benchmark outperforms every learned model, learned skill concentrates entirely in clustered periods, and the residual unsolved problem, namely isolated, non-clustered moderate events, is localised to specific seismicity regimes characterised here. The task addressed throughout is precursor pattern detection, the statistical association of pre-event seismicity patterns with impending moderate earthquakes; no claim of deterministic prediction is made or implied.

2. Study region and data

2.1. Tectonic setting

Three coupled tectonic domains fall inside the study window (10–30°N, 85–98°E; Fig. 1a). Along the eastern Himalayan front, the Main Himalayan Thrust takes up roughly 15–20 mm/yr of convergence and has repeatedly hosted great ruptures. South of it, the Shillong Plateau behaves as a crustal pop-up whose bounding faults produced the 1897 earthquake; Bilham and England [2] argued from that event for seismogenic strain partitioning well away from the main front. The third domain, and the one that turns out to matter most for our results, is the Indo-Burman wedge, where Indian lithosphere descends obliquely beneath Myanmar under several kilometres of Bengal-fan sediment. Seismicity there splits into shallow wedge activity and a persistent band of intermediate-depth intraslab events near 30–60 km; geodesy indicates the shallow megathrust is locked and accumulating strain beneath some of the most densely populated terrain on Earth [3,4]. A detection framework for this region therefore has to cope with thrust, strike-slip, and deep intraslab populations at once, which is precisely what makes it a demanding testbed.

2.2. Catalog acquisition and cleaning

We queried the USGS National Earthquake Information Center ComCat service for the study polygon over 1 January 1970 to 31 December 2025 and kept only reviewed events of type earthquake. Records lacking any hypocentral parameter were dropped, as were the handful outside plausible physical bounds (magnitude 1.0–10.0, depth 0–700 km) and exact duplicates on the time–location–magnitude tuple. What remains is a catalog of 7,188 events spanning 19 February 1970 to 29 December 2025, magnitudes 2.9–7.7, depths 0.6–200 km. No magnitude homogenisation across reported scales was attempted; at the $M \geq 4.5$ target level the catalog is dominated by teleseismically constrained body- and moment-magnitude solutions, and we prefer transparency over conversion assumptions. Fig. 1 shows the rate history and magnitude distribution; the rate step visible in the early-to-mid 1990s motivates the completeness treatment that follows.

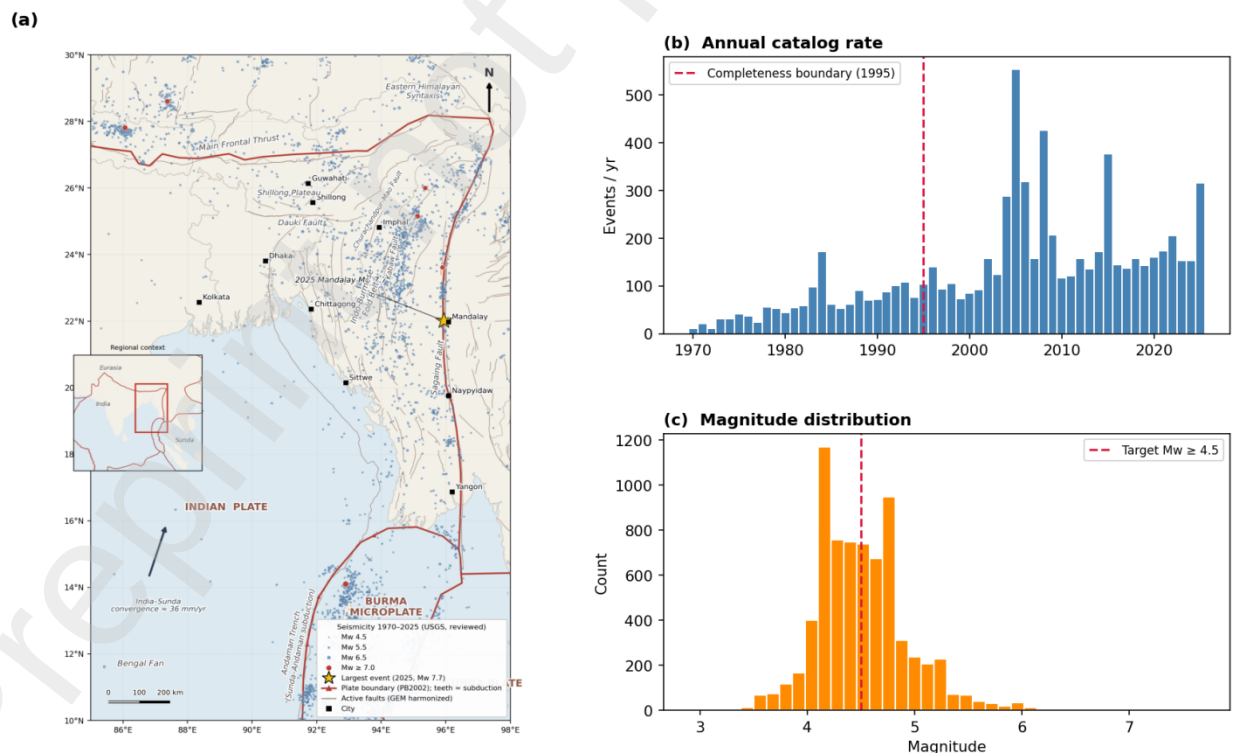


Fig. 1. Study region and catalog. (a) The Eastern Indian Plate margin in EPSG:4326: PB2002 plate boundaries [19] with subduction teeth along the Sunda–Andaman trench, harmonized active faults of the GEM Global Active Faults Database [20], and reviewed USGS seismicity 1970–2025 ($M_w \geq 4.5$; symbols graduated by magnitude; red: $M_w \geq 7.0$; star: the 2025 Mandalay $M_w 7.7$, the largest catalogued event); inset: regional plate context. (b) Annual event rate with the 1995 completeness-era boundary. (c) Magnitude distribution with the $M_w \geq 4.5$ target threshold.

2.3. Catalog completeness and the dual-protocol design

Detection capability in the region changed drastically over the study period: the median annual rate is 51 events/yr before 1990 but 156 events/yr after 2000, with the sustained increase coinciding with the completion of modern global digital seismographic coverage in the early-to-mid 1990s. Because several features used here are explicit or implicit rate measures, such non-stationarity would otherwise be absorbed into the classifier as spurious signal [13,14]. Two protocols are therefore analysed in parallel. Protocol A uses the full 1970–2025 catalog and serves as a robustness and ablation reference. Protocol B, the primary protocol, restricts analysis to sequences ending on or after 1 January 1995, retaining 5,697 sequences of which 1,302 (22.85%) are positive. The boundary was fixed from the rate history alone, before any model comparison, following the completeness-era stratification logic long established in statistical seismology.

2.4. Micro-sequence construction and labelling

The catalog was transformed into overlapping micro-sequences of $N = 20$ consecutive events with stride 1, yielding 7,168 sequences, each summarised by its centroid location and end time. A sequence is labelled positive when at least one $M_w \geq 4.5$ event occurs strictly after its end time, within 15 days, and within 100 km of its centroid (great-circle distance); otherwise it is labelled background. This produces 1,504 positive sequences (20.98%; imbalance 1:3.77). The 15-day horizon and 100-km radius define an operationally meaningful short-term, local alerting problem; the magnitude threshold corresponds to events of engineering relevance on the soft-sediment sites that dominate the region. Event-driven windowing keeps per-sequence sample size constant across activity regimes at the cost of variable window duration, which the feature set addresses through inter-event-time statistics. We flag at the outset a property of this labelling that Section 4.5 makes quantitative: because moderate events cluster, a large share of positives sit inside ongoing sequences, so any faithful evaluation must include a clustering benchmark.

3. Methodology

3.1. Physics-informed feature suite

Each sequence is described by thirteen features (Table 1). Gutenberg–Richter parameters [21] use the maximum-likelihood b -value [22] with the standard a -value companion. Cumulative Benioff strain (CBS) is computed from the standard energy relation, and its acceleration is quantified by a bounded convexity measure: the normalised CBS trajectory is integrated over real elapsed time and referenced to 0.5, so that positive values indicate late-window strain-release acceleration in the spirit of accelerating moment release [16,17,23] while remaining bounded and free of the power-law fitting instabilities that have drawn criticism of classical AMR analysis [18,24]. An analogous arrival-convexity feature, α_{micro} , applies the same construction to the normalised event-arrival curve and measures acceleration of event occurrence itself. Depth structure is captured by mean focal depth and depth variance; spatial organisation by an events-per-area density ρ and a latitude-binned spatial entropy; magnitude structure by the window mean and the

maximum-minus-mean prominence; and temporal structure by the mean inter-event time, its reciprocal rate λ , and a robust quartile-based coefficient of dispersion CV_t that is insensitive to single quiescence gaps.

Table 1. Feature definitions and precursor/background discriminability over all 7,168 sequences (Mann–Whitney U, two-sided). Point-biserial correlation r is given with respect to the binary label.

Feature	Physical meaning	Prec. mean	Backg. mean	p-value	r
b	G–R b-value (MLE)	0.806	0.750	4.8×10^{-9}	+0.089
a	G–R a-value	4.501	4.234	3.8×10^{-12}	+0.097
CBS_accel	Benioff-strain convexity	0.057	0.015	3.2×10^{-29}	+0.122
α_micro	Arrival-curve convexity	0.054	0.025	1.3×10^{-30}	+0.114
σ^2_z	Depth variance (km ²)	364.8	1033.4	$< 10^{-300}$	−0.459
mean_z	Mean focal depth (km)	27.98	44.64	5.7×10^{-302}	−0.465
ρ	Spatial density (ev/km ²)	1.9×10^{-3}	3×10^{-5}	$< 10^{-300}$	+0.366
H_space	Spatial entropy	1.122	1.212	4.4×10^{-6}	−0.138
mean_M	Mean magnitude	4.535	4.503	6.1×10^{-6}	+0.071
M_max	Max-minus-mean magnitude	0.911	0.976	1.6×10^{-11}	−0.058
λ	Event rate (1/day)	37.75	0.544	$< 10^{-300}$	+0.421
mean_dt	Mean inter-event time (d)	0.925	3.302	$< 10^{-300}$	−0.341
CV_t	Robust dispersion of Δt	2.202	3.278	5.3×10^{-17}	−0.033

Four features (ρ , σ^2_z , λ , mean_dt) separate the classes at significance levels below double-precision underflow and are reported as $p < 10^{-300}$ rather than as literal zeros. Fig. 2 shows the univariate discriminability ranking.

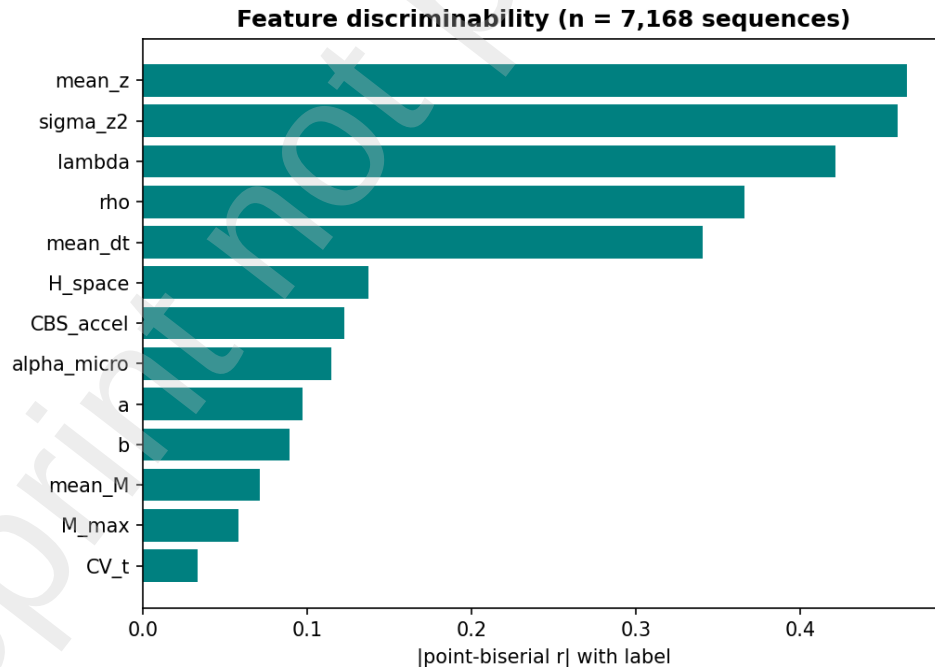


Fig. 2. Absolute point-biserial correlation between each feature and the precursor label ($n = 7,168$).

3.2. Classifier

The classifier is a random forest of 300 trees with balanced-subsample class weighting and a minimum leaf size of two [25], chosen for its robustness to heterogeneous feature scales, native handling of nonlinear interactions, and interpretable impurity-based importance. To suppress seed-level variance, the alert score is the average of the predicted precursor probability over three forests with fixed seeds {42, 7, 2026}. Synthetic oversampling was deliberately avoided: interpolation-based schemes such as SMOTE synthesise implausible seismicity states and, under temporal validation, risk manufacturing train–test similarity; class weighting achieves the required cost sensitivity without such artefacts. A probability-calibration variant (Platt scaling for training folds below 2,000 samples, isotonic regression otherwise [26,27]) was evaluated and is reported as an ablation in Section 4.5.

3.3. Embargoed time-series cross-validation

Model assessment uses ten-fold forward-chaining time-series cross-validation in which each fold trains strictly on the past and tests on the subsequent block. Two leakage channels remain in the naive version of this scheme. First, consecutive sequences share 19 of 20 events, so training sequences immediately preceding a test block are near-duplicates of its first members. Second, labels look 15 days into the future, so training labels adjacent to the boundary encode test-period events. Both channels are closed by an embargo gap of 20 sequences (approximately 56 days at the median inter-event spacing) between every training set and its test block, directly implementing the purged and embargoed cross-validation logic of López de Prado [12]. Folds whose training set contains fewer than 2,500 sequences are additionally flagged as under-trained in fold-level reporting. The quantitative effect of the embargo is reported as an ablation.

3.4. Fuzzy alerting and Pareto-knee threshold selection

The averaged forest score is converted to a graded alert membership through a fuzzy sigmoid $\mu_{\text{alert}}(P) = 1/(1 + \exp(-\lambda_f (P - \theta^*)))$ with sharpness $\lambda_f = 10$ [28], acknowledging that alert relevance is inherently graded rather than crisp. The operating threshold θ^* is selected by sweeping 297 candidate thresholds over the pooled out-of-fold scores, computing recall and false alarm rate $\text{FAR} = \text{FP}/(\text{FP} + \text{TN})$ at each, and choosing the knee of the resulting empirical trade-off front as the point minimising the Euclidean distance to the ideal (recall = 1, FAR = 0). This scalarised two-objective selection replaces the arbitrary 0.5 cut-off with an explicitly cost-aware operating point; the label "multi-objective" refers to this bi-objective trade-off analysis, which is solved here by exhaustive one-dimensional sweep rather than by a population-based evolutionary algorithm, a distinction stated plainly to avoid overclaiming (cf. [29]). Because θ^* is selected on pooled out-of-fold scores, the reported operating-point metrics carry a mild selection optimism, which is bounded by reporting bootstrap confidence intervals and the full threshold sweep.

3.5. Physical plausibility gate family

Alert consumers rarely tolerate the false-alarm budget that a knee point implies, so we asked whether a physically motivated veto could buy precision cheaply. The obvious construction, and the one most often proposed, is a hard conjunction: gate G1 admits an alert only if both α_{micro} and p exceed the median of the background training distribution in the current fold, with the two thresholds recomputed from training data alone so the gate itself cannot leak. This pair, comprising one temporal acceleration measure and one spatial concentration measure, was chosen a priori, before any gated result had been examined. Because a

conjunction of two median-referenced tests is severe by construction, we also built two softer variants. G2 lets alerts whose score already exceeds a bypass level θ_h skip the physical test entirely, applying it only to borderline alerts with $\theta^* \leq P < \theta_h$; sweeping θ_h then traces a continuous branch between the hard-gated and ungated extremes, so the gate becomes a dial rather than a filter. G3 replaces the crisp conjunction with a product-geometric T-norm over sigmoid memberships of the same two features and thresholds the combined score. Section 5.4 dissects why the hard version fails and what that failure teaches about mixing learned scores with rule-based physics.

3.6. Baselines, benchmark, metrics, and uncertainty

Skill numbers mean little without reference points, so four are evaluated under the identical embargoed folds. Two are learned: an L2-regularised logistic regression on standardised features with balanced class weights, and an equal-weight score average of forest and regression. One is deliberately trivial: the raw window event-rate feature λ used alone as a ranking score. The fourth, and the one we regard as the benchmark any pattern claim must clear, is a parameter-free Omori persistence kernel: each sequence is scored by $\sum_j (\Delta t_j + c)^{-1}$ over all catalogued $M \geq 4.5$ events within 100 km of its centroid during the preceding 365 days, with Δt_j the elapsed days and $c = 0.05$ d [30,31]. This score encodes nothing but the empirical fact that moderate earthquakes follow one another in space-time clusters; it requires no training, so it is immune to every leakage channel by construction, and a homogeneous Poisson model (the natural null) corresponds to a constant score with ROC-AUC 0.5. Discrimination is summarised by ROC-AUC and, given imbalance, PR-AUC against stated prevalence; probabilistic quality by the Brier score; operating points by recall, precision, FAR, and F1 with 95% bootstrap intervals from 1,000 resamples. Overlapping windows autocorrelate the out-of-fold sample, so these intervals should be read as lower bounds on uncertainty; a block-bootstrap treatment is identified as follow-up work.

3.7. Algorithm summary

Algorithm 1 assembles the full detection pipeline in the order executed by the released code; every manuscript number derives from one pass of this procedure per protocol.

Algorithm 1. Fuzzy Pareto-thresholded random-forest precursor detection with embargoed temporal validation and clustering benchmark.

```

Input : catalog C (7,188 events, time-sorted); window N=20;
        label triple (M*=4.5, T=15 d, R=100 km); era year Y=1995;
        folds K=10; embargo g=20; seeds S={42,7,2026};
        threshold grid  $\theta$  (297 pts); gate percentile q=50
Output: alert score p(s); operating point  $\theta^*$ ; gate families G1, G2

Phase 1: Sequence representation
1: for i = 1 ... |C|-N do
2:   s_i  $\leftarrow$  events e_{i...e_{i+N-1}}; centroid c_i; end time t_i
3:   x_i  $\leftarrow$  13 features (Table 1)
4:   y_i  $\leftarrow$  1 if  $\exists$  event M  $\geq M^*$  in (t_i, t_i+T] within R of c_i

Phase 2: Completeness protocols
5: D_B  $\leftarrow$  {s_i : year(t_i)  $\geq$  Y} (primary); D_A  $\leftarrow$  all s_i

Phase 3: Embargoed forward-chaining CV

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6: for k = 1 ... K do
7:   (Train_k, Test_k) ← time-ordered split with
      max(Train_k) + g < min(Test_k)
8:   for  $\sigma \in S$ : fit RF_ $\sigma$ (300, balanced-subsample, leaf  $\geq 2$ )
      on Train_k
9:    $p(s) \leftarrow \text{mean}_{\sigma} \text{RF}_{\sigma}(s)$  for all  $s \in \text{Test}_k$ 
10:   $\delta_{\alpha}, \delta_{\rho} \leftarrow q$ -th pct of background features, Train_k only

Phase 4: Pareto-knee threshold selection
11: for  $\theta \in \Theta$ : Recall( $\theta$ ), FAR( $\theta$ ) on pooled out-of-fold p
12:  $\theta^* \leftarrow \text{argmin}_{\theta} \sqrt{[1 - \text{Recall}(\theta)]^2 + \text{FAR}(\theta)^2}$ 
13:  $\mu_{\text{alert}}(s) \leftarrow 1 / (1 + \exp(-\lambda_f (p(s) - \theta^*)))$ ,  $\lambda_f = 10$ 

Phase 5: Physical plausibility gates
14:  $G1(s) \leftarrow [p \geq \theta^*] \wedge [\alpha_{\text{micro}} > \delta_{\alpha}] \wedge [\rho > \delta_{\rho}]$ 
15:  $G2(s; \theta_h) \leftarrow [p \geq \theta_h] \vee G1(s)$ ; sweep  $\theta_h \in [\theta^*, 0.95]$ 

Phase 6: Benchmark and reporting
16:  $p_{\Omega}(s) \leftarrow \Sigma_j (\Delta t_j + c)^{-1}$  over  $M \geq M^*$  events within R,
      past 365 d (no training; leakage-free by construction)
17: report pooled metrics + 95% bootstrap CIs + per-fold
      tables; flag folds with  $|\text{Train}_k| < 2,500$ 

```

4. Results

4.1. Global discrimination

Table 2 summarises both protocols. Under the primary completeness-aware Protocol B (5,170 out-of-fold sequences, prevalence 24.8%), the multi-seed forest attains ROC-AUC 0.892, PR-AUC 0.836, and Brier score 0.091. The full-catalog Protocol A yields ROC-AUC 0.868 and PR-AUC 0.799 over 6,510 out-of-fold sequences (prevalence 21.4%), confirming that catalog-era heterogeneity depresses measurable skill. Fig. 3 shows the corresponding ROC and precision–recall curves.

Table 2. Global discrimination and Pareto-knee operating points under embargoed 10-fold time-series cross-validation. CIs are 95% bootstrap intervals.

Quantity	Protocol B (post-1995, primary)	Protocol A (full catalog)
Out-of-fold sequences (prevalence)	5,170 (24.8%)	6,510 (21.4%)
ROC-AUC	0.892	0.868
PR-AUC	0.836	0.799
Brier score	0.091	0.082
Knee threshold θ^*	0.229	0.186
Recall at θ^*	0.796 (0.773–0.817)	0.753 (0.731–0.773)
Precision at θ^*	0.730 (0.708–0.752)	0.626 (0.602–0.648)
FAR at θ^*	0.098 (0.088–0.107)	0.122 (0.113–0.131)
F1 at θ^*	0.761	0.684
Confusion counts (TP/FN/FP/TN)	1022 / 262 / 379 / 3507	1047 / 343 / 625 / 4495

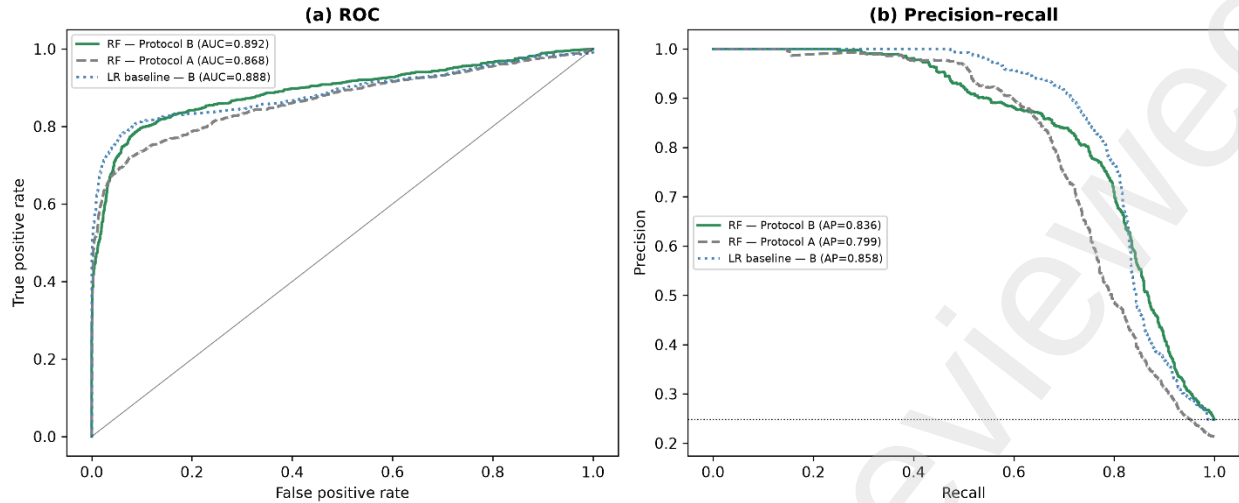


Fig. 3. (a) ROC and (b) precision–recall curves for the random forest under both protocols, with the logistic-regression baseline under Protocol B. Dotted horizontal line: Protocol B prevalence.

4.2. Operating-point selection

Fig. 4 traces the 297-point sweep for Protocol B. The knee criterion lands at $\theta^* = 0.229$, where recall reaches 0.796 with precision 0.730 and FAR 0.098; the F1 optimum sits close by, so the knee is not an artefact of the particular distance measure. It is worth pausing on what the conventional 0.5 cut-off would have delivered instead: recall 0.576 at FAR 0.024. Moving to the knee purchases about 280 additional true precursor windows at the cost of a manageable rise in false alerts. Whether that trade is attractive depends on the consumer, but wherever a missed moderate event costs more than several tolerable false alerts, as is typical for infrastructure operators on amplifying soft-sediment sites, the knee is the defensible default, and the sweep itself gives any other consumer the full menu.

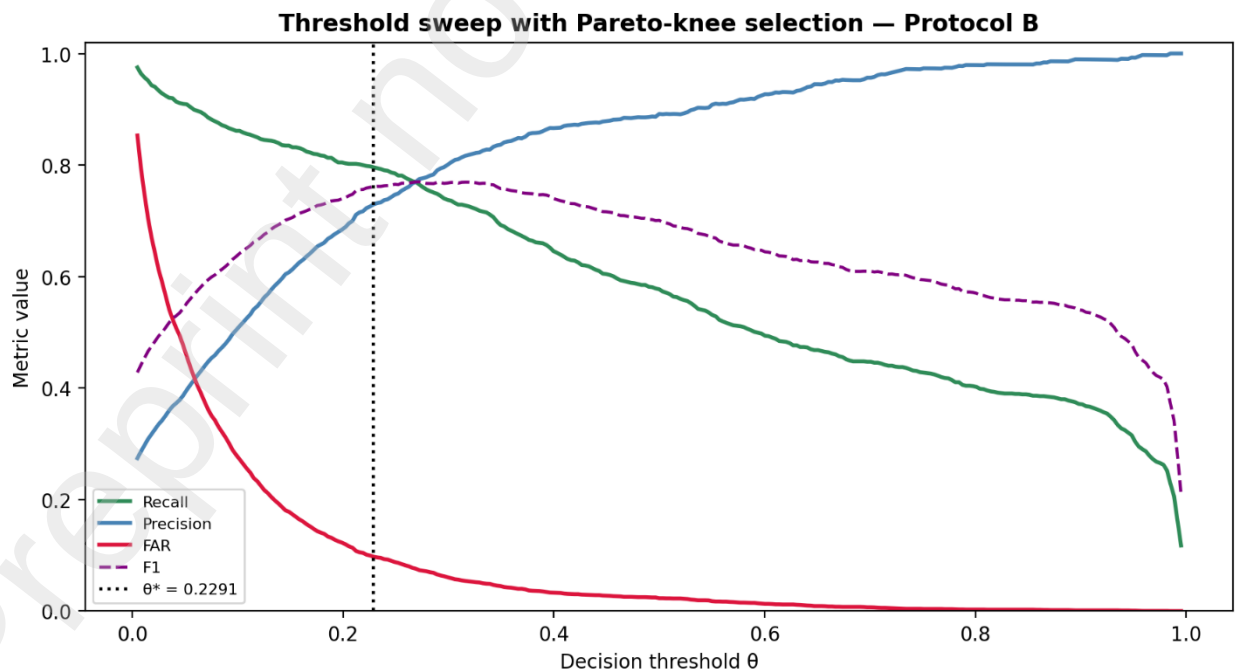


Fig. 4. Threshold sweep of recall, precision, FAR, and F1 for Protocol B with the Pareto-knee threshold $\theta^* = 0.229$.

4.3. Fold-level temporal stability

Pooled numbers flatter this system, and Table 3 shows how. Five of the ten Protocol B test blocks are excellent, with recall between 0.84 and 0.98. Fold 1 detects none of its 20 positives, unsurprising given only 507 training sequences. The interesting failures are folds 6 and 8, which collapse to recall 0.179 and 0.093 while sitting on training sets of 3,092 and 4,126 sequences respectively. Training-set size, in other words, explains the Protocol A instability pattern (qualified folds recover to 0.773 ± 0.157 there) but not the Protocol B failures, where the qualified subset is actually worse (0.556 ± 0.350) than the full set (0.610 ± 0.380). Something other than data volume is at work in those two windows. Table 6 characterises them, and Section 5.3 argues they are the most informative result in the paper.

Table 3. Per-fold performance at $\theta^* = 0.229$ under Protocol B. Folds with training size below 2,500 sequences are marked with †.

Fold	n_train	Test positives	Recall	Precision	FAR	F1
1†	507	20	0.000	0.000	0.012	0.000
2†	1,024	397	0.937	0.892	0.375	0.914
3†	1,541	187	0.850	0.768	0.145	0.807
4†	2,058	93	0.978	0.659	0.111	0.788
5	2,575	104	0.837	0.569	0.160	0.677
6	3,092	39	0.179	0.233	0.048	0.203
7	3,609	164	0.963	0.806	0.108	0.878
8	4,126	54	0.093	0.106	0.091	0.099
9	4,643	85	0.624	0.602	0.081	0.613
10	5,160	141	0.638	0.756	0.077	0.692

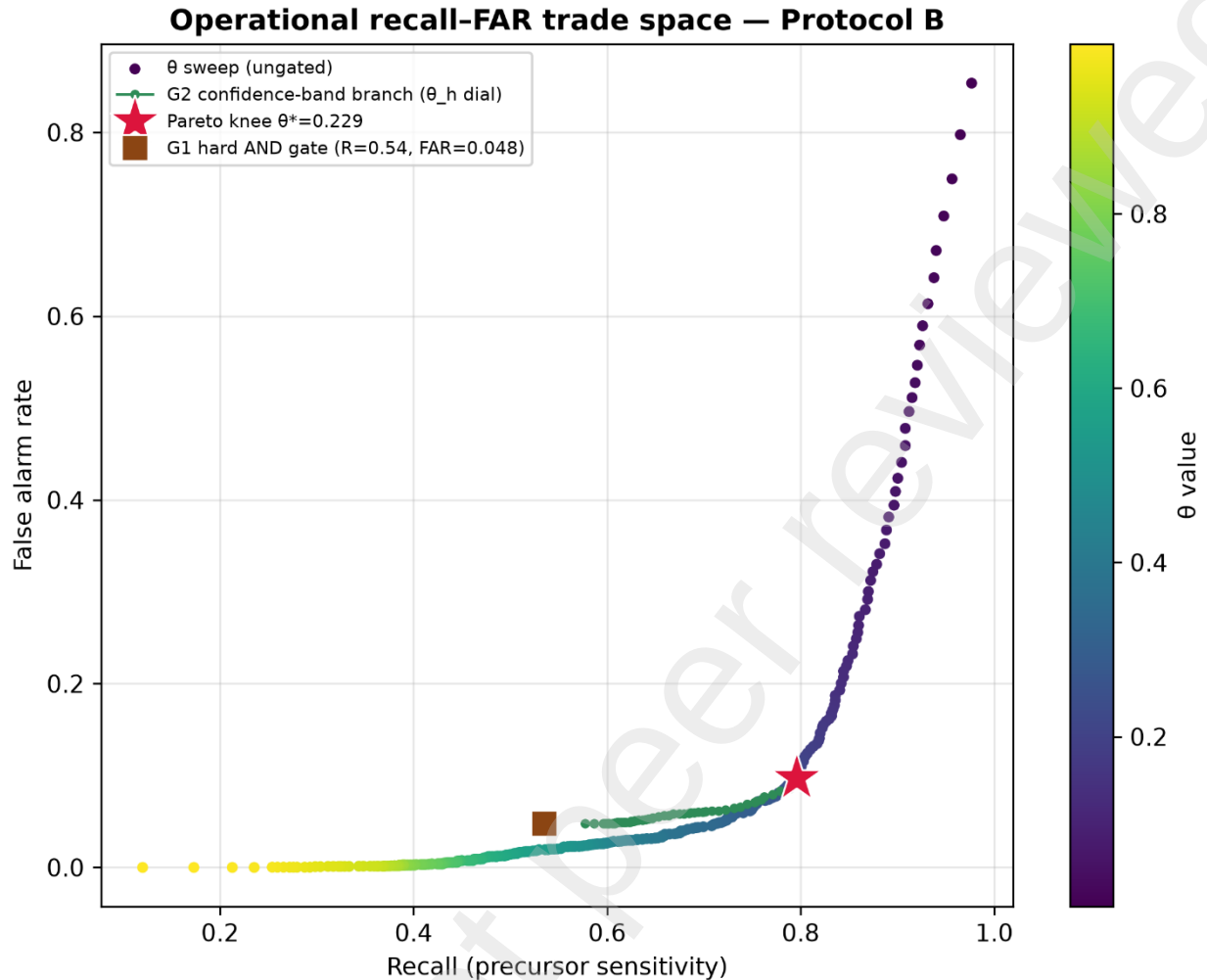


Fig. 6. Per-fold recall and FAR at the protocol-specific knee threshold. Shaded folds train on fewer than 2,500 sequences.

4.4. Gating

Table 4 and Fig. 5 report the gate family for Protocol B, and the outcome is blunt. The hard conjunction G1 does what it promises on false alarms, cutting FAR from 0.098 to 0.048, but pays with about 335 discarded true alerts and a recall of 0.535; under the knee criterion it is dominated and never selected. G2 collapses to the ungated knee under the same criterion, which is itself informative: once the score is well ordered, unconditional vetoing removes more information than it contributes. What survives of G2 is its dial property: sweeping the bypass level θ_h traces the continuous branch in Fig. 5 between (recall 0.796, FAR 0.098) and the hard-gated extreme, so an operator with a fixed false-alarm budget can buy it at a transparent, quantified recall price. The fuzzy T-norm variant G3 is Pareto-dominated throughout and is retained only for completeness. Why the hard gate fails so badly, given that both gate features are strongly label-associated, turns out to have a clean quantitative answer, developed in Section 5.4.

Table 4. Alert configurations under Protocol B (pooled out-of-fold, $n = 5,170$).

Configuration	Recall	Precision	FAR	F1
Random forest, $\theta = 0.50$	0.576	0.889	0.024	0.699
Pareto-knee, $\theta^* = 0.229$	0.796	0.729	0.098	0.761
G1 hard AND gate	0.535	0.788	0.048	0.637
G2 confidence-band (knee-selected)	0.796	0.729	0.098	0.761
G3 fuzzy T-norm ($g^* = 0.452$)	0.766	0.652	0.135	0.704

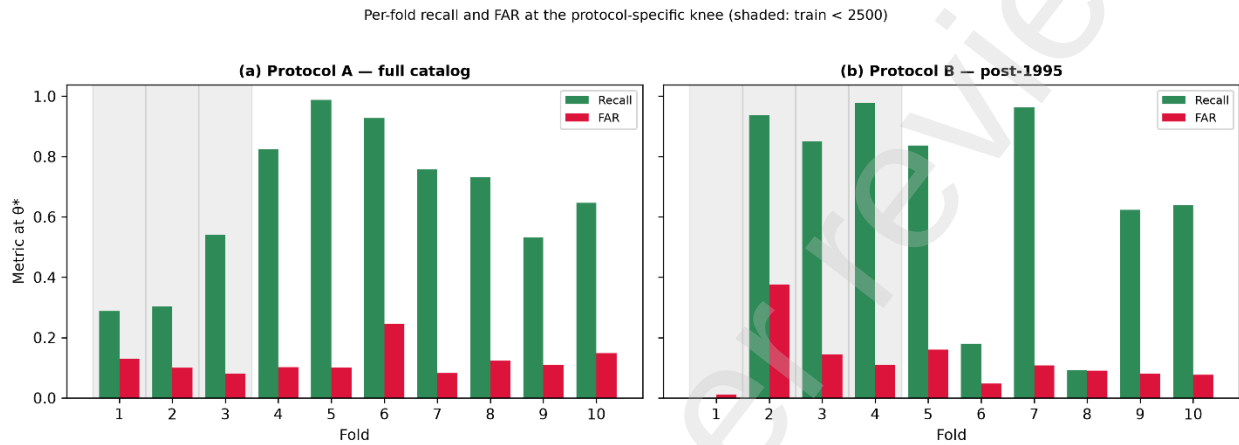


Fig. 5. Operational recall–FAR trade space for Protocol B: ungated threshold sweep (coloured by θ), Pareto knee (star), hard gate G1 (square), and the continuous G2 confidence-band branch (green).

4.5. Baselines, the Omori benchmark, and ablations

Table 5 is where the headline changes. Logistic regression on the same features attains ROC-AUC 0.888 and PR-AUC 0.858, statistically comparable to the forest; the equal-weight forest–regression ensemble edges both (0.897/0.873); and the raw event-rate feature alone reaches ROC-AUC 0.899, statistically indistinguishable from the forest itself. The parameter-free Omori persistence kernel, by contrast, uses no features, no training, and only a single line of physics, yet attains ROC-AUC 0.933 and PR-AUC 0.890 on the identical out-of-fold sequences, and at its own knee delivers recall 0.866 at FAR 0.083, outperforming every learned configuration on every headline metric. A rank-average of forest and kernel (0.930) does not improve on the kernel alone, so the learned models contribute no measurable information beyond clustering persistence. Stratification makes the point sharper: within the low-clustering half of the sample (kernel score below the median, prevalence 3.5%), the forest's ROC-AUC is 0.494, a value indistinguishable from chance, while within the high-clustering half it is 0.913. The remaining ablations behave as expected: probability calibration degrades both ranking (0.875) and, counter-intuitively, Brier score (0.112 versus 0.091), a known failure mode of isotonic regression on temporally heterogeneous folds, and removing the embargo inflates ROC-AUC under the primary protocol (0.896 versus 0.892), a small but systematic optimism.

Table 5. Model comparison, clustering benchmark, and ablations under Protocol B (embargoed unless stated). Dashes denote not-applicable entries.

Model / benchmark / ablation	ROC-AUC	PR-AUC	Knee recall	Knee FAR
Random forest (3-seed, primary)	0.892	0.836	0.796	0.098
Logistic regression	0.888	0.858	0.812	0.093
Single-feature λ baseline	0.899	0.863	—	—
RF + LR ensemble	0.897	0.873	0.822	0.098
Omori persistence kernel (benchmark)	0.933	0.890	0.866	0.083
Rank ensemble RF + Omori	0.930	0.902	—	—
RF with probability calibration	0.875	0.789	0.786	0.122
RF without embargo (gap = 0)	0.896	0.842	—	—

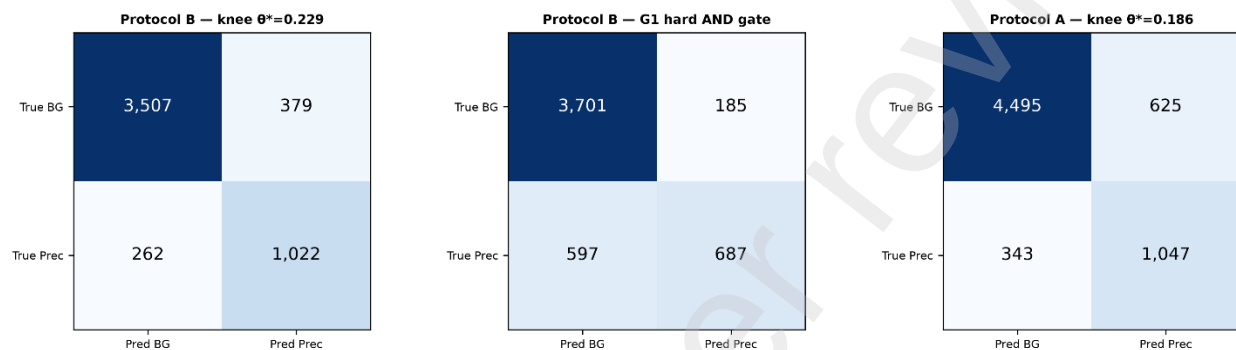


Fig. 7. Confusion matrices at the knee operating points (Protocol B ungated and hard-gated; Protocol A ungated).

4.6. Feature importance

Impurity-based importance averaged over the ten Protocol B folds (Fig. 8) concentrates on the clustering and rate block (λ , ρ , mean inter-event time, depth variance, and mean depth), with the convexity features contributing weak but consistently positive increments, in line with their univariate effect sizes. Read together with Table 5, the importance profile is no longer a list of "precursor signals": it is a description of how the forest reconstructs clustering persistence from window statistics.

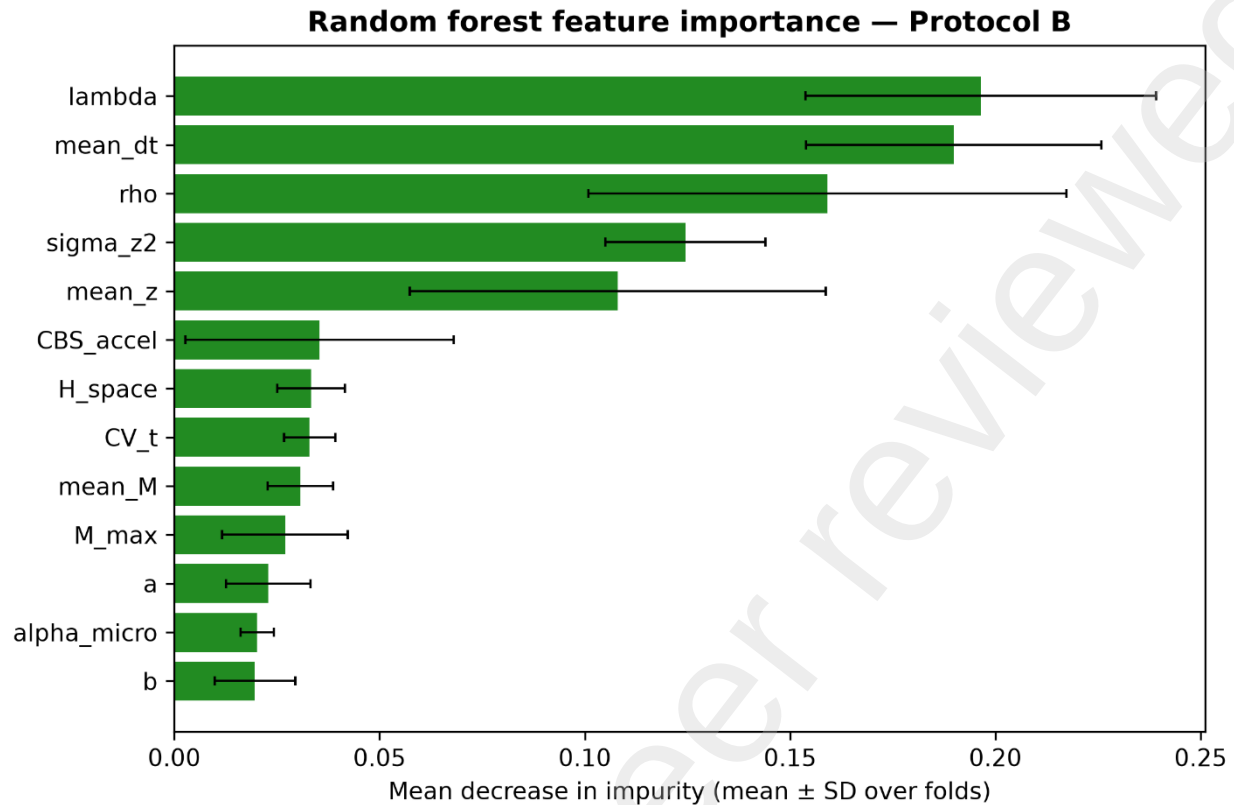


Fig. 8. Mean-decrease-in-impurity feature importance (mean $\pm$ SD over ten Protocol B folds).

4.7. Event-level case studies and audit

Per-sequence metrics answer the question "is this window elevated?", but the consumer of an alerting system asks a different one: what did the system say before the earthquakes that mattered? We audited all 26 mainshocks of $M_w \geq 6.2$ falling inside the pseudo-prospective coverage (March 2000 onward), tracing every out-of-fold sequence whose centroid lay within 100 km of the epicentre during the preceding 60 days. Because the scores are out-of-fold, each trace shows what the deployed system would have displayed at the time. Table 7 and Fig. 9 summarise the audit, and the 26 events sort cleanly into three behaviours.

The first behaviour is success, and it is instructive about what success means here. Before the 12 May 2015 M_w 7.3 Dolakha earthquake, every nearby sequence in the 15-day window was in alert (maximum score 1.000, with every in-window sequence above threshold), and the 26 April M_w 6.7 and 12 May M_w 6.3 events show the same saturated picture. All three are aftershocks of the 25 April Gorkha mainshock, whose eastern aftershock zone crosses into the study box: the alerts began when the sequence began, and the normalised Omori trace in Fig. 9 follows the forest score almost point for point. The second behaviour is the quiet miss. Before the 24 August 2016 M_w 6.8 Chauk earthquake (82 km deep, Indo-Burman slab) sequences existed nearby, yet the maximum score reached 0.086; this event gives the fold-6/8 population of Table 6 a name and a date, and the 2021 M_w 6.2 Hakha event repeats the anatomy. The third behaviour is the one the sequence-level tables cannot show at all: for twenty of the 26 mainshocks, including the 28 March 2025 Mandalay M_w 7.7, the 2009 Andaman M_w 7.5, the 2011 Sikkim M_w 6.9, and the 2016 Manipur M_w 6.7, no sequence centroid lay within 100 km at any point in the prior 60 days. Their epicentral

regions were locally quiescent, and under event-driven windowing a quiescent region produces no evaluable object: these earthquakes were not missed so much as structurally invisible. The per-mainshock scorecard is therefore roughly four detections out of 26, every one a continuation of an ongoing aftershock sequence, and it should be read alongside, rather than in place of, the sequence-level recall of 0.796 that is dominated by Mw 4.5–5.5 cluster members.

Table 7. Event-level audit of $M_w \geq 6.2$ mainshocks in the pseudo-prospective period. "Seq." counts out-of-fold sequences with centroid within 100 km in the prior 60/15 days; score and alert fraction refer to the 15-day window at $\theta^ = 0.229$. Dashes denote not-applicable entries.*

Date	Mw	z (km)	Context	Seq. 60/15 d	Max score	Alerted	Outcome
2015-04-26	6.7	23	Gorkha aftershock zone	56 / 56	0.999	100%	Alerted
2015-05-12	7.3	15	Gorkha aftershock zone	87 / 13	1.000	100%	Alerted
2015-05-12	6.3	15	Gorkha aftershock zone	92 / 18	1.000	100%	Alerted
2004-12-26	6.2	13	Sumatra–Andaman aftershock	5 / 5	0.326	100%	Alerted
2004-12-26	6.3	26	Sumatra–Andaman aftershock	2 / 2	0.197	0%	Missed
2016-08-24	6.8	82	Indo-Burman intraslab (Chauk)	6 / 5	0.086	0%	Missed at score ≈ 0.08
2021-11-25	6.2	43	Indo-Burman intraslab (Hakha)	3 / 0	—	—	Missed; no 15-d coverage
2008-08-10	6.2	20	Andaman ridge	87 / 0	—	—	Local activity ceased 44 d before
2025-03-28	7.7	10	Sagaing fault (Mandalay)	0 / 0	—	—	No local sequence: invisible
2009-08-10	7.5	24	Andaman	0 / 0	—	—	No local sequence: invisible
2011-09-18	6.9	50	Sikkim Himalaya	0 / 0	—	—	No local sequence: invisible
2016-01-03	6.7	55	Manipur intraslab	0 / 0	—	—	No local sequence: invisible
(+14 events)	6.2–6.9	—	Various	0 / 0	—	—	No local sequence: invisible

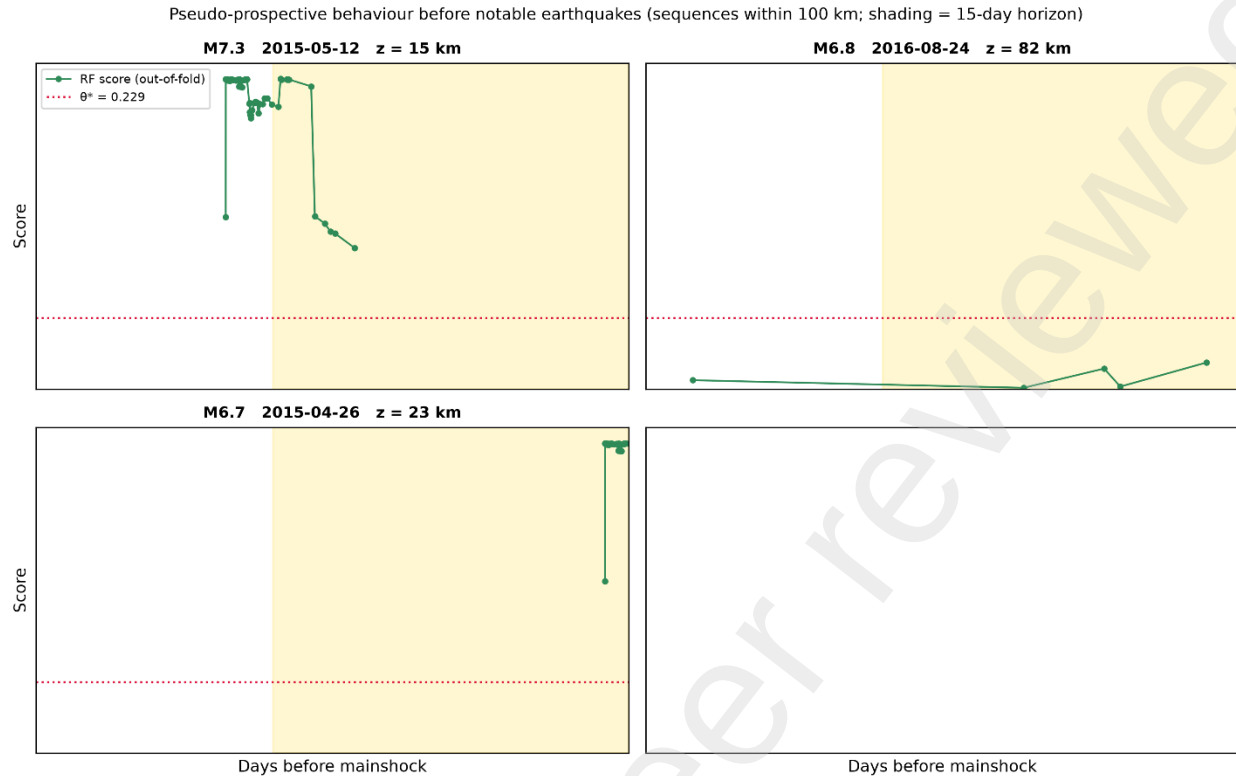


Fig. 9. Pseudo-prospective behaviour before four notable earthquakes: out-of-fold forest score (green), normalised Omori kernel (dashed), knee threshold (dotted), and the 15-day label horizon (shaded). Top row and lower left: Gorkha aftershock events, saturated alerts driven by the ongoing sequence. Lower right: the 2016 Chauk Mw 6.8 intraslab event, scored near zero throughout.

The three behaviours invite a systematic question: across every target event, how often does each occur, and why? We therefore extended the audit to all 2,363 $M_w \geq 4.5$ events falling within the pseudo-prospective period, classifying each into one of three outcomes: alerted (at least one out-of-fold sequence within 100 km in the prior 15 days scored at or above θ^*), covered but sub-threshold, and no coverage (no local sequence existed to score). Table 8 and Fig. 10 give the taxonomy, and the headline pairing is deliberately double-edged. Conditional on coverage the system performs well: 84.3% of covered events were preceded by an alert. Unconditionally it detects 18.6% of target events, because 78.0% of events had no local sequence to evaluate. The stratifications locate the failure structure precisely. Detection decays monotonically with depth (26.7% below 35 km, 6.0% at 35–70 km, and exactly zero for all 355 events deeper than 70 km) and does not increase with magnitude: Mw 5.5–5.9 events are detected less often (12.0%) than Mw 4.5–4.9 events (18.9%). Detection probability, in other words, is set by whether recent shallow micro-activity happened to surround the future epicentre, not by how large the coming event is. That is what a clustering detector looks like when audited at the event level, and we suggest that per-event audits of this kind, reported alongside sequence-level metrics, should become standard practice in this literature.

Table 8. Outcome taxonomy for all 2,363 $M_w \geq 4.5$ events in the pseudo-prospective period. Conditional detection is the alerted fraction among covered events.

Stratum	n	Alerted	Covered, sub- θ^*	No coverage	Cond. detection
All events	2,363	18.6%	3.5%	78.0%	84.3%
Mw 4.5–4.9	1,789	18.9%	3.6%	77.5%	84.1%
Mw 5.0–5.4	429	19.3%	2.8%	77.9%	87.4%
Mw 5.5–5.9	108	12.0%	3.7%	84.3%	76.5%
Mw 6.0+	37	13.5%	5.4%	81.1%	71.4%
Depth < 35 km	1,538	26.7%	1.6%	71.7%	94.3%
Depth 35–70 km	470	6.0%	5.5%	88.5%	51.9%
Depth > 70 km	355	0.0%	8.7%	91.3%	0.0%

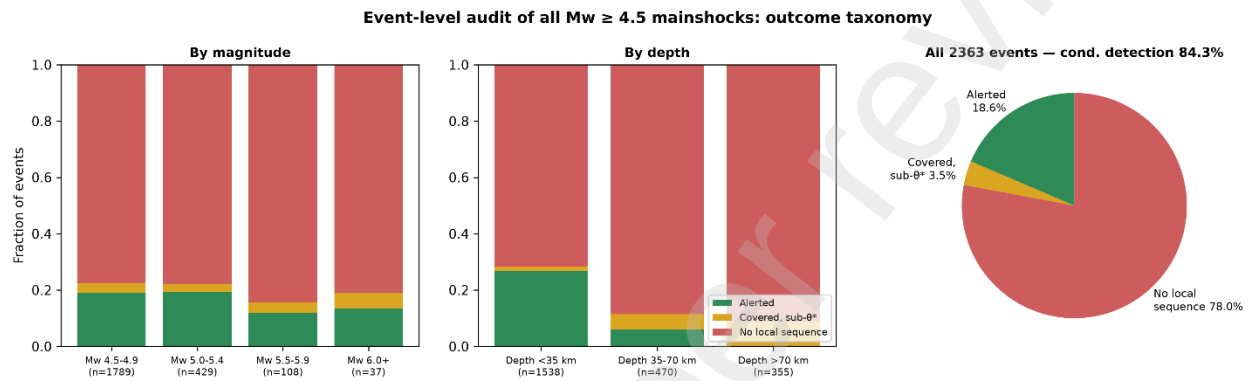


Fig. 10. Event-level outcome taxonomy for all Mw ≥ 4.5 mainshocks, stratified by magnitude and depth, with the overall outcome distribution. Green: alerted; ochre: covered but sub-threshold; red: no local sequence to score.

5. Discussion

5.1. What the detector has actually learned

Before the benchmark, the importance profile invited a comfortable reading: shallow, compact, accelerating micro-clusters as an elevated-hazard state, the classical foreshock-like picture. The benchmark forces a harder one. A kernel that knows nothing except when and where the last few moderate events happened outranks the forest, adding the forest to that kernel adds nothing, and the forest is blind exactly where clustering is absent. The parsimonious conclusion is that the forest has learned to be an Omori law, expressed in the vocabulary of window statistics. We do not regard this as a failure of the model so much as a diagnosis of the task: with positives defined as any $M \geq 4.5$ within 15 days and 100 km, most positive windows sit inside ongoing sequences, and clustering persistence is the signal, the situation that classical aftershock-hazard statistics describe [32]. The result echoes, with a concrete regional quantification, the meta-analytic finding that simple models match elaborate ones in this literature [6], and it is precisely the kind of null that modern forecast-evaluation practice exists to expose [15,33]. Any study reporting skill on a comparable task without a clustering benchmark should, we suggest, be read as unfinished.

5.2. The clustering-persistence ceiling

The stratified analysis bounds what feature engineering can achieve here. Within clustered periods the ordering problem is nearly solved by the kernel itself, leaving little headroom; outside them the forest performs at chance, and its two catastrophic folds are exactly the windows where the kernel also collapses

(recall 0.205 and 0.093 in folds 6 and 8). Whatever information would be needed to anticipate non-clustered moderate events, it is not present in this catalog's window statistics at this feature resolution, or it does not exist at all. Distinguishing those two possibilities is now a well-posed question, which is more than could be said before the benchmark was run.

5.3. Anatomy of the failing windows

Table 6 characterises each Protocol B test window. The two failures are not random: both are multi-year quiescent intervals over the Indo-Burman wedge (fold 6: October 2011 to April 2015; fold 8: March 2017 to September 2020) whose positives are deep and isolated. Their triggering events sit at mean depths near 40 km against 25–27 km for training-era positives, their window event rates are two orders of magnitude below training-era positive rates ($\lambda \approx 0.5$ – 0.6 per day versus ≈ 36 – 46), and their spatial densities are lower by a similar factor. These are intermediate-depth intraslab earthquakes arriving without any foreshock-like build-up, concentrated near 21°N , 94°E . The forest assigns their windows mean scores of 0.07–0.09: the forest has never seen positives that resemble these, because in the training past positives of this type barely exist. The 1995 completeness boundary is not implicated: the failing windows are recent, well-recorded, and their negatives are classified correctly (FAR stays at 0.07–0.10). What the boundary could not fix is that the region's seismicity alternates between clustered shallow-wedge regimes and quiet intervals punctuated by isolated deep events, and only the first regime is learnable from clustering statistics. For the seismological reader, this localises the genuinely open problem in space, depth, and time: short-term anticipation of non-clustered intermediate-depth Indo-Burman intraslab events.

Table 6. Character of each Protocol B test window. Positive-class means are shown for depth, rate, and density; RF and Omori recall are at their respective pooled knees.

Fold	Test window	Prev.	Pos. depth (km)	Pos. λ (/d)	Pos. ρ (ev/km ²)	RF recall	Omori recall
1	2000-03 – 2004-07	0.04	36.7	4.6	1.6×10^{-2}	0.000	0.650
2	2004-07 – 2005-04	0.77	31.3	15.7	8.4×10^{-5}	0.937	0.947
3	2005-04 – 2006-09	0.36	22.8	83.5	2.4×10^{-3}	0.850	0.888
4	2006-09 – 2008-10	0.18	13.6	53.7	1.8×10^{-3}	0.978	0.978
5	2008-10 – 2011-10	0.20	25.9	20.0	4.4×10^{-3}	0.837	0.817
6	2011-10 – 2015-04	0.08	40.7	0.6	1.5×10^{-5}	0.179	0.205
7	2015-04 – 2017-03	0.32	12.6	104.1	2.9×10^{-3}	0.963	0.951
8	2017-03 – 2020-09	0.10	38.1	0.5	1.2×10^{-5}	0.093	0.093
9	2020-09 – 2023-08	0.16	22.9	29.1	3.8×10^{-3}	0.624	0.647
10	2023-08 – 2025-12	0.27	22.9	40.4	1.5×10^{-3}	0.638	0.688

Fold 2 deserves its own footnote: it spans the months after the December 2004 great Sumatra–Andaman rupture at the southern edge of the study box, its prevalence of 0.77 reflects the ensuing aftershock flood, and its FAR of 0.375 is computed on only 120 negatives, a reminder that per-fold ratios in extreme-prevalence windows are volatile even when the ordering is good. Fold 1 pairs low prevalence with an under-trained model and is uninformative about the architecture; the kernel, needing no training, recalls 0.800 there.

5.4. Why hard physical gating fails

The gate's failure invites two explanations. Either the forest had already internalised the physical constraints, making the gate redundant, or the gate features lack discriminative power where it matters. The data support a sharper, compound account. Decomposing the 1,439 knee alerts by gate outcome shows that the 557 gate-rejected alerts were correct 62.1% of the time, two and a half times the 24.8% base rate, so the gate was discarding alerts that were mostly right; redundancy alone cannot explain destroying that much signal. Ranking the same alerts by each gate feature against residual truth separates the two culprits: within alerts, α_{micro} carries essentially no residual information (AUC 0.524), whereas ρ retains a great deal (AUC 0.904). The conjunction therefore behaves as an informative test AND-ed with a coin flip, and a coin flip in a conjunction only removes alerts. Three design lessons follow. A feature already supplied to the classifier should not be re-imposed as a crisp veto, because the score has consumed its marginal signal and what remains after thresholding on the score is residual, not marginal, information. Any conjunction is only as good as its weakest member, and the weakest member here was one of our own novel convexity features, an honest cost of reporting it. And where residual information does survive in a physical variable, as it does for ρ , the productive architecture is soft reweighting or stacking of the score, not a veto; the G2 dial approximates this and correspondingly dominates G1 throughout Fig. 5. We suspect that this pattern, in which hard physical filters underperform the learned scores they police for identifiable information-theoretic reasons, generalises well beyond seismology, and quantifying the within-alert residual AUC of each gate feature, as done here, is a two-line diagnostic any hybrid system can run.

5.5. Operational reading

For an agency wanting something deployable tomorrow, the practical summary is almost embarrassingly simple: run the Omori kernel. It needs the catalog, one line of arithmetic, no retraining, and it delivers recall 0.866 at FAR 0.083 under evaluation conditions designed to be hostile. The learned stack earns its keep not as a better detector but as the instrument that established what the kernel ceiling is, where it fails, and what any future model must exceed before claiming precursory content. At the forest's knee the alert duty fraction is 27.8% of sequence-time, and the G2 dial supplies intermediate false-alarm budgets down to about 0.05 at quantified recall cost. We caution against cross-study comparisons of these figures with the regional ML literature evaluated under random cross-validation; alarm-based skill measures on common ground, in the Molchan tradition, are the right currency for that conversation [34,35].

5.6. Limitations

Seven limitations bound the claims, and we prefer to state them plainly. The study is single-region and single-catalog, so nothing here establishes transportability. Overlapping windows autocorrelate the out-of-fold sample; the embargo protects the train-test boundary, but the bootstrap intervals treat interior points as independent and are therefore optimistic, pending a block-bootstrap or disjoint-stride treatment. The knee threshold is chosen on pooled out-of-fold scores, importing mild optimism into the headline operating point; the released sweep tables let any reader apply a nested selection instead. The Omori kernel, while decisive as a benchmark, is not a full ETAS model, since it lacks magnitude-dependent productivity and spatial kernels, and a fitted ETAS comparison remains the formal next step [15,36]; we note, however, that a stronger clustering model can only widen, never close, the gap the kernel has already opened over the learned detectors. Labels are binary and magnitude-agnostic above the threshold. The labelling triple itself

($M_w \geq 4.5$, 15 d, 100 km) has not yet been subjected to a sensitivity grid; given how thoroughly clustering dominates at this setting, the grid is better read as mapping where, if anywhere, non-clustering signal emerges, than as robustness confirmation. Finally, and exposed only by the event-level audit, event-driven windowing constrains coverage itself: sequences exist only where the last twenty regional events occurred, so 78% of target events, and effectively all deep ones, present no local object to score. Complementing event-driven windows with fixed-duration spatial cells (for example, 90-day windows on a half-degree grid) is the structural remedy; it changes the sampling frame rather than the model, and no feature engineering inside the present frame can substitute for it.

6. Future research directions

The verified pipeline and its negative results define a concrete programme. (i) ETAS-referenced evaluation: fitting a full epidemic-type aftershock sequence model to the identical catalog and scoring it on identical alarm windows would replace the parameter-free kernel with the strongest available clustering null, sharpening the boundary of genuinely precursory claims [15,36]. (ii) Genuine evolutionary design: the labelling triple, decision threshold, and gate parameters span a mixed design space in which a population-based NSGA-II search maximising hypervolume on the recall–FAR–alert-duty front would replace the present one-dimensional sweep and make the "multi-objective" designation literal [29]; run across the label-sensitivity grid, it doubles as the systematic search for label settings where learned models escape the clustering ceiling. (iii) Targeting the residual problem: folds 6 and 8 define a concrete adversarial subset of non-clustered, intermediate-depth Indo-Burman events, for which slab-specific covariates (slab geometry, intraslab stress indicators, regional GNSS coupling [4]) and deep sequence models that represent regime switching are the natural candidates; neural temporal point processes such as RECAST remove hand-crafted windowing entirely [37], and simulation-trained nowcasting transformers offer a complementary route [38]. (iv) Prospective evaluation under community standards: freezing this pipeline, kernel included, for registered CSEP-style testing with Molchan-diagram reporting would convert retrospective association into auditable forecast skill [35,39,40]. (v) Physics-informed enrichment: static Coulomb stress transfer from recent moderate events [41,42], InSAR-derived deformation, and transient-robust magnitude statistics such as the b-positive estimator [43] are directly injectable as sequence covariates, and the stratified benchmark is now available to test whether any of them adds skill outside clusters. (vi) Enhanced catalogs and transfer: deep-learning phase pickers can lower the regional completeness magnitude and may populate exactly the quiescent windows that defeat all present methods [44,45], while Himalaya-to-Indo-Burman domain-adaptation experiments would quantify regime transportability inside one framework. (vii) Decision-grade uncertainty: block-bootstrap and conformal wrappers would attach honest finite-sample guarantees to alert sets [46], and cost-curve elicitation with regional civil-protection agencies would replace the geometric knee with an explicitly economic operating point.

7. Conclusions

We set out to build a short-term precursor detector for $M_w \geq 4.5$ earthquakes in the Eastern Indian Plate margin that a hostile reviewer could not fault on validation grounds, and we largely succeeded: the embargoed, completeness-aware random forest reaches ROC-AUC 0.892 with recall 0.796 at a 9.8% false

alarm rate, every number regenerable from seeded code. The result we did not set out to find is the important one. A parameter-free Omori persistence kernel, evaluated on the same folds, beats every learned configuration (ROC-AUC 0.933, recall 0.866 at FAR 0.083); no ensemble recovers an advantage; the forest is at chance outside clustered periods; and the two windows that defeat it, quiescent intervals whose positives are deep, isolated Indo-Burman intraslab events, defeat the kernel identically. The honest statement of the state of the art on this task is therefore: clustering persistence is the detectable signal, a one-line physical kernel is the operational tool of choice, the learned framework's value is diagnostic rather than predictive, and the open scientific problem has been localised to a specific tectonic population rather than left as a pooled-metric abstraction. An audit of all 2,363 target events sharpens the operational truth further: alerts preceded 84.3% of the events the system could see, but it could see only one event in five, and none below 70 km, a constraint of the windowing frame itself rather than of any model within it. We also showed, quantitatively, why bolting a hard physical veto onto a learned score destroys skill (a crisp conjunction containing one residually uninformative feature can only delete correct alerts) and offered the within-alert residual AUC as a routine diagnostic for hybrid designs. The framework, benchmark, ablations, and failure anatomy together set a reproducible floor that ETAS-referenced, prospectively tested successors must clear, and we release everything needed to hold them, and us, to it.

Data and code availability

The earthquake catalog is publicly available from the USGS ComCat service. The complete pipeline, comprising catalog cleaning, sequence construction, feature extraction, labelling, quality checking, the final modelling script, the Omori benchmark, and all figure-generation code, is provided with fixed random seeds, together with all intermediate tables (out-of-fold predictions including benchmark scores, threshold sweeps, per-fold metrics, and gate comparisons), permitting bit-level verification of every reported number. [An archived DOI, e.g. via Zenodo, will be added at submission.]

CRedit authorship contribution statement

Emran Islam: Conceptualization, Methodology, Software, Data curation, Investigation, Writing – original draft, Project administration. Samsun Nahar: Methodology (fuzzy inference and membership-function design), Conceptualization, Writing – review and editing. Mariam Sultana: Formal analysis, Visualization, Writing – review and editing. Abdul Halim Bhuiyan: Validation, Supervision, Writing – review and editing.

Declaration of generative AI in scientific writing

During the preparation of this work the authors used a large-language-model assistant (Claude) to support computational verification of the analysis pipeline, statistical benchmarking, and language editing of the manuscript. The authors reviewed and edited all content and takes full responsibility for the content of the published article.

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] Oldham RD. Report on the great earthquake of 12th June 1897. *Mem Geol Surv India* 1899;29:1–379.
- [2] Bilham R, England P. Plateau ‘pop-up’ in the great 1897 Assam earthquake. *Nature* 2001;410:806–9. <https://doi.org/10.1038/35071057>
- [3] Kayal JR. Microearthquake seismology and seismotectonics of South Asia. Dordrecht: Springer; 2008. <https://doi.org/10.1007/978-1-4020-8180-4>
- [4] Steckler MS, Mondal DR, Akhter SH, Seeber L, Feng L, Gale J, et al. Locked and loading megathrust linked to active subduction beneath the Indo-Burman Ranges. *Nat Geosci* 2016;9:615–8. <https://doi.org/10.1038/ngeo2760>
- [5] DeVries PMR, Viégas F, Wattenberg M, Meade BJ. Deep learning of aftershock patterns following large earthquakes. *Nature* 2018;560:632–4. <https://doi.org/10.1038/s41586-018-0438-y>
- [6] Mignan A, Broccardo M. Neural network applications in earthquake prediction (1994–2019): meta-analytic and statistical insights on their limitations. *Seismol Res Lett* 2020;91(4):2330–42. <https://doi.org/10.1785/0220200021>
- [7] Beroza GC, Segou M, Mousavi SM. Machine learning and earthquake forecasting—next steps. *Nat Commun* 2021;12:4761. <https://doi.org/10.1038/s41467-021-24952-6>
- [8] Mousavi SM, Beroza GC. Deep-learning seismology. *Science* 2022;377:eabm4470. <https://doi.org/10.1126/science.abm4470>
- [9] Panakkat A, Adeli H. Neural network models for earthquake magnitude prediction using multiple seismicity indicators. *Int J Neural Syst* 2007;17(1):13–33. <https://doi.org/10.1142/S0129065707000890>
- [10] Asim KM, Martínez-Álvarez F, Basit A, Iqbal T. Earthquake magnitude prediction in Hindukush region using machine learning techniques. *Nat Hazards* 2017;85:471–86. <https://doi.org/10.1007/s11069-016-2579-3>
- [11] Asencio-Cortés G, Martínez-Álvarez F, Troncoso A, Morales-Esteban A. Medium–large earthquake magnitude prediction in Tokyo with artificial neural networks. *Neural Comput Appl* 2017;28(5):1043–55. <https://doi.org/10.1007/s00521-015-2121-7>
- [12] López de Prado M. *Advances in financial machine learning*. Hoboken: Wiley; 2018. <https://doi.org/10.1002/9781119482086>
- [13] Wiemer S, Wyss M. Minimum magnitude of completeness in earthquake catalogs: examples from Alaska, the western United States, and Japan. *Bull Seismol Soc Am* 2000;90(4):859–69. <https://doi.org/10.1785/0119990114>
- [14] Woessner J, Wiemer S. Assessing the quality of earthquake catalogues: estimating the magnitude of completeness and its uncertainty. *Bull Seismol Soc Am* 2005;95(2):684–98. <https://doi.org/10.1785/0120040007>
- [15] Mizrahi L, Dallo I, van der Elst NJ, et al. Developing, testing, and communicating earthquake forecasts: current practices and future directions. *Rev Geophys* 2024;62(3):e2023RG000823. <https://doi.org/10.1029/2023RG000823>
- [16] Bufe CG, Varnes DJ. Predictive modeling of the seismic cycle of the greater San Francisco Bay region. *J Geophys Res* 1993;98(B6):9871–83. <https://doi.org/10.1029/93JB00357>
- [17] Bowman DD, Ouillon G, Sammis CG, Sornette A, Sornette D. An observational test of the critical earthquake concept. *J Geophys Res* 1998;103(B10):24359–72. <https://doi.org/10.1029/98JB00792>
- [18] Mignan A. Retrospective on the Accelerating Seismic Release (ASR) hypothesis: controversy and new horizons. *Tectonophysics* 2011;505(1–4):1–16. <https://doi.org/10.1016/j.tecto.2011.03.010>
- [19] Bird P. An updated digital model of plate boundaries. *Geochem Geophys Geosyst* 2003;4(3):1027. <https://doi.org/10.1029/2001GC000252>
- [20] Styron R, Pagani M. The GEM Global Active Faults Database. *Earthq Spectra* 2020;36(1_suppl):160–80. <https://doi.org/10.1177/8755293020944182>
- [21] Gutenberg B, Richter CF. Frequency of earthquakes in California. *Bull Seismol Soc Am* 1944;34(4):185–8. <https://doi.org/10.1785/BSSA0340040185>

- [22] Aki K. Maximum likelihood estimate of b in the formula $\log N = a - bM$ and its confidence limits. *Bull Earthq Res Inst Univ Tokyo* 1965;43:237–9.
- [23] Benioff H. Earthquakes and rock creep. *Bull Seismol Soc Am* 1951;41(1):31–62. <https://doi.org/10.1785/BSSA0410010031>
- [24] Jaumé SC, Sykes LR. Evolving towards a critical point: a review of accelerating seismic moment/energy release prior to large and great earthquakes. *Pure Appl Geophys* 1999;155:279–305. <https://doi.org/10.1007/s000240050266>
- [25] Breiman L. Random forests. *Mach Learn* 2001;45:5–32. <https://doi.org/10.1023/A:1010933404324>
- [26] Platt JC. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In: *Advances in large margin classifiers*. Cambridge, MA: MIT Press; 1999, p. 61–74.
- [27] Niculescu-Mizil A, Caruana R. Predicting good probabilities with supervised learning. In: *Proceedings of the 22nd international conference on machine learning*; 2005, p. 625–32. <https://doi.org/10.1145/1102351.1102430>
- [28] Zadeh LA. Fuzzy sets. *Inf Control* 1965;8(3):338–53. [https://doi.org/10.1016/S0019-9958\(65\)90241-X](https://doi.org/10.1016/S0019-9958(65)90241-X)
- [29] Deb K, Pratap A, Agarwal S, Meyarivan T. A fast and elitist multiobjective genetic algorithm: NSGA-II. *IEEE Trans Evol Comput* 2002;6(2):182–97. <https://doi.org/10.1109/4235.996017>
- [30] Omori F. On the after-shocks of earthquakes. *J Coll Sci Imp Univ Tokyo* 1894;7:111–200.
- [31] Utsu T. A statistical study on the occurrence of aftershocks. *Geophys Mag* 1961;30:521–605.
- [32] Reasenber PA, Jones LM. Earthquake hazard after a mainshock in California. *Science* 1989;243:1173–6. <https://doi.org/10.1126/science.243.4895.1173>
- [33] Hardebeck JL, Felzer KR, Michael AJ. Improved tests reveal that the accelerating moment release hypothesis is statistically insignificant. *J Geophys Res* 2008;113:B08310. <https://doi.org/10.1029/2007JB005410>
- [34] Molchan GM. Strategies in strong earthquake prediction. *Phys Earth Planet Inter* 1990;61(1–2):84–98. [https://doi.org/10.1016/0031-9201\(90\)90097-H](https://doi.org/10.1016/0031-9201(90)90097-H)
- [35] Zechar JD, Jordan TH. Testing alarm-based earthquake predictions. *Geophys J Int* 2008;172(2):715–24. <https://doi.org/10.1111/j.1365-246X.2007.03676.x>
- [36] Ogata Y. Statistical models for earthquake occurrences and residual analysis for point processes. *J Am Stat Assoc* 1988;83(401):9–27. <https://doi.org/10.1080/01621459.1988.10478560>
- [37] Dascher-Cousineau K, Shchur O, Brodsky EE, Günnemann S. Using deep learning for flexible and scalable earthquake forecasting. *Geophys Res Lett* 2023;50:e2023GL103909. <https://doi.org/10.1029/2023GL103909>
- [38] Rundle JB, Turcotte DL, Donnellan A, Grant Ludwig L, Luginbuhl M, Gong G. Nowcasting earthquakes. *Earth Space Sci* 2016;3(11):480–6. <https://doi.org/10.1002/2016EA000185>
- [39] Schorlemmer D, Werner MJ, Marzocchi W, et al. The Collaboratory for the Study of Earthquake Predictability: achievements and priorities. *Seismol Res Lett* 2018;89(4):1305–13. <https://doi.org/10.1785/0220180053>
- [40] Jordan TH, Chen Y-T, Gasparini P, Madariaga R, Main I, Marzocchi W, et al. Operational earthquake forecasting: state of knowledge and guidelines for utilization. *Ann Geophys* 2011;54(4):315–91. <https://doi.org/10.4401/ag-5350>
- [41] King GCP, Stein RS, Lin J. Static stress changes and the triggering of earthquakes. *Bull Seismol Soc Am* 1994;84(3):935–53. <https://doi.org/10.1785/BSSA0840030935>
- [42] Steacy S, Gombert J, Cocco M. Introduction to special section: stress transfer, earthquake triggering, and time-dependent seismic hazard. *J Geophys Res* 2005;110:B05S01. <https://doi.org/10.1029/2005JB003692>
- [43] van der Elst NJ. B-positive: a robust estimator of aftershock magnitude distribution in transiently incomplete catalogs. *J Geophys Res Solid Earth* 2021;126:e2020JB021027. <https://doi.org/10.1029/2020JB021027>
- [44] Zhu W, Beroza GC. PhaseNet: a deep-neural-network-based seismic arrival-time picking method. *Geophys J Int* 2019;216(1):261–73. <https://doi.org/10.1093/gji/ggy423>
- [45] Mousavi SM, Ellsworth WL, Zhu W, Chuang LY, Beroza GC. Earthquake transformer—an attentive deep-learning model for simultaneous earthquake detection and phase picking. *Nat Commun* 2020;11:3952. <https://doi.org/10.1038/s41467-020-17591-w>
- [46] Angelopoulos AN, Bates S. Conformal prediction: a gentle introduction. *Found Trends Mach Learn* 2023;16(4):494–591. <https://doi.org/10.1561/22000000101>